STAFF REVIEW · H446 254 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. Cycle 6 (=4+3−1).

2. The instruction being transferred from memory.

3. The CU identifies and interprets the opcode/operands.

4. 1+0+1=2 decimal = 10_2, so Sum=0 and Carry-out=1.

I Do — source worked steps

1. Predict first — The instruction fetch and data read may both need the same memory transfer pathway.

2. Explain and check — The requests may have to be scheduled rather than served simultaneously; this is related to the Von Neumann bottleneck.

We Do — guided CHECK

1. Predict first
Separate program and data memories indicate Harvard-style organisation.
Explain and check
Independent transfer pathways can allow concurrent instruction fetch and data access, subject to the processor design.

You Do — source mark schemes

1. [4] Von Neumann uses a shared instruction/data memory space [2]; Harvard has separate instruction/data memories or address spaces [2].

2. [6] Shared transfer pathway [1] used for both instructions and data [1], which can cause contention [1]; Harvard uses independent paths [1], potentially overlapping transfers [1] although overall speed depends on implementation [1].

3. [6] Unified main memory is shared [1], a Von Neumann feature [1]; separate L1 caches [1] offer separate instruction/data access paths [1], a Harvard-like feature [1]; this is a hybrid or modified Harvard organisation [1].

Exit Ticket — source answer

1. False. Hybrid designs are common.